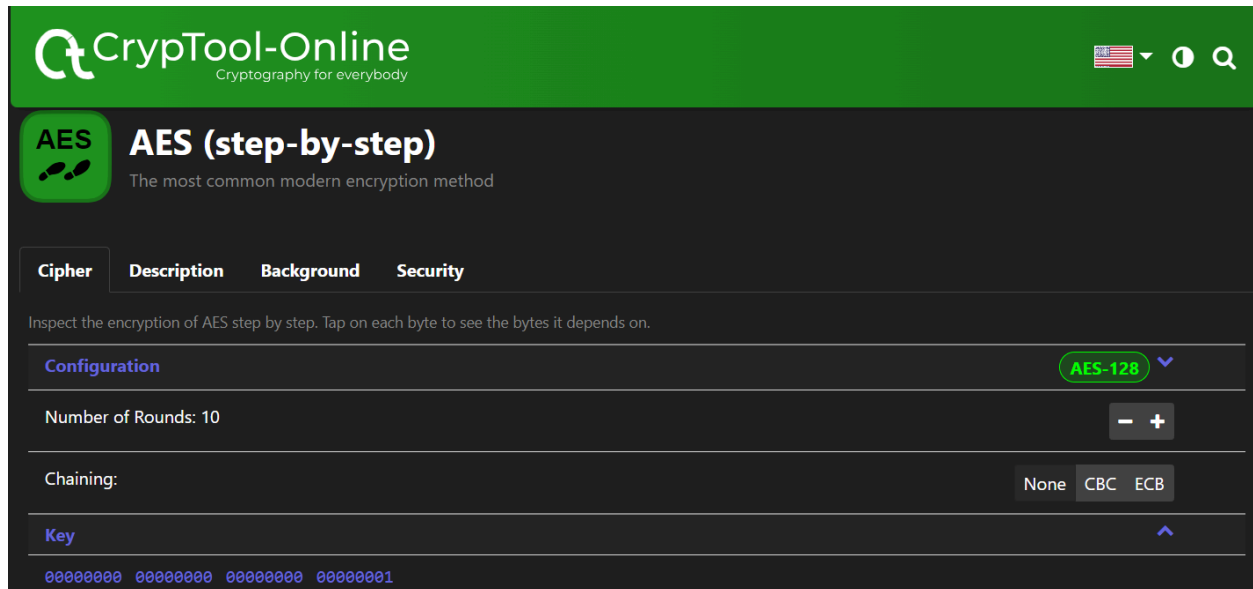
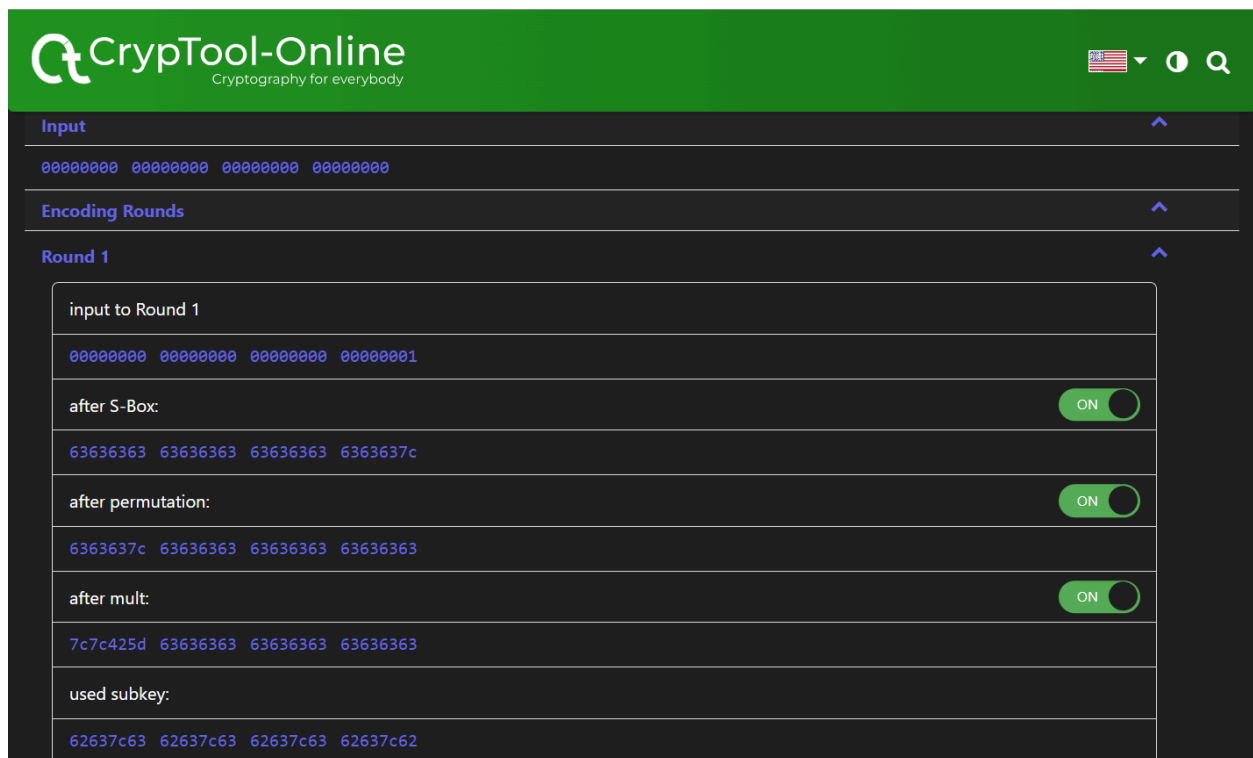


TASK1: Understanding AES encryption.



The above screenshot displays the basic details like the key and input that was given.



Comparison between plain text and round 1 of AES encryption.

CrypTool-Online
Cryptography for everybody



Round 9

Round 10

input to Round 10

dd9da703 9ca1f158 42434694 5dc8687a

after S-Box:

ON

c15e5c7b de32a16a 2c1a5a22 4ce845da

after permutation:

ON

c1325ada de1a457b 2ce85c6a 4c5ea122

used subkey:

c477f00f b3b8ec07 1a8b8d29 6663bda6

after mix with key:

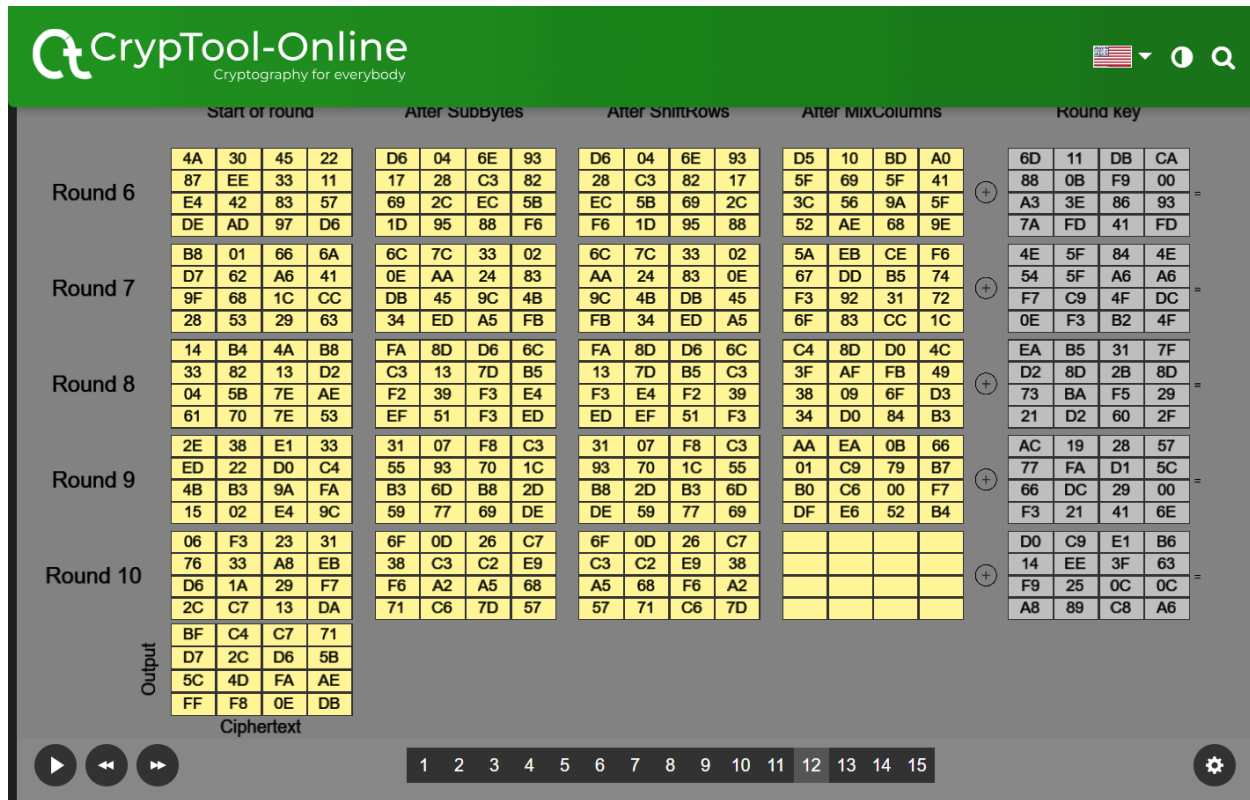
ON

0545aad5 6da2a97c 3663d143 2a3d1c84

Encoded

0545aad5 6da2a97c 3663d143 2a3d1c84

The final round of encryption and the encoded output.



Visualization of AES encryption(animation).

TASK2: Wi-Fi Authentication Crack

```
crypto@crypto:~$ sudo apt-get install aircrack-ng
[sudo] password for crypto:
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package aircrack-ng
crypto@crypto:~$ man aircrack-ng
crypto@crypto:~$
```

Commands used for Installing aircrack-ng and going through its manual pages.

```
crypto@crypto:~/Documents$ aircrack-ng wep_64_ptw.cap
Opening wep_64_ptw.cap
Read 65282 packets.
```

#	BSSID	ESSID	Encryption
1	00:12:BF:12:32:29	Appart	WEP (30566 IVs)

Choosing first network as target.

Opening wep_64_ptw.cap
Attack will be restarted every 5000 captured ivs.
Starting PTW attack with 30566 ivs.

Aircrack-ng 1.2 rc4

[00:00:00] Tested 1514 keys (got 30566 IVs)

KB	depth	byte(vote)
0	0/ 9	1F(39680) 4E(38400) 14(37376) 5C(37376) 9D(37376)
1	7/ 9	64(36608) 3E(36352) 34(36096) 46(36096) BA(36096)
2	0/ 1	1F(46592) 6E(38400) 81(37376) 79(36864) AD(36864)
3	0/ 3	1F(40960) 15(38656) 7B(38400) BB(37888) 5C(37632)
4	0/ 7	1F(39168) 23(38144) 97(37120) 59(36608) 13(36352)

KEY FOUND! [1F:1F:1F:1F:1F]

Decrypted correctly: 100%

Crack the WEP password using the PTW mod

```
crypto@crypto:~/Documents$ aircrack-ng -K wep_KoreK.ivs
Opening wep_KoreK.ivs
Read 567298 packets.
```

#	BSSID	ESSID	Encryption
1	00:11:95:91:78:8C		WEP (566693 IVs)

Choosing first network as target.

Opening wep_KoreK.ivs
Reading packets, please wait...

Aircrack-ng 1.2 rc4

[00:00:11] Tested 1995 keys (got 566693 IVs)

KB	depth	byte(vote)
0	0/ 1	AE(50) 11(20) 71(20) 0D(12) 10(12) 68(12) 84(12) 0A(9) 31(6) 90(6) 9D(6) 83(5)
1	1/ 2	5B(31) BD(18) F8(17) E6(16) 35(15) 7A(13) 7F(13) 81(13) CF(13) D2(13) 29(12) 58(12)
2	0/ 3	7F(31) 74(24) 54(17) 1C(13) 73(13) 86(12) 1B(10) BF(10) 31(8) 56(8) 5C(8) FF(8)
3	0/ 1	3A(148) EC(20) EB(16) FB(13) 81(12) D7(12) ED(12) F0(12) E9(12) F8(10) DD(9) 02(8)
4	0/ 1	03(140) 90(31) 4A(15) 8F(14) E9(13) AD(12) 86(10) DB(10) E2(10) 99(8) 59(6) 93(6)
5	0/ 1	D0(69) 04(27) 60(24) C8(24) 26(20) A1(20) A0(18) 4F(17) B6(16) 69(14) 84(14) A3(13)
6	0/ 1	AF(124) D4(29) C8(20) EE(18) 3F(12) 54(12) 3C(11) 90(11) 76(10) CF(10) 3D(9) 3E(9)
7	0/ 1	9B(168) 90(24) 72(22) F5(21) 11(20) F1(20) 86(17) FB(16) 0E(15) 12(13) AD(13) 17(12)
8	0/ 1	F6(157) EE(24) 66(20) DA(18) E0(18) EA(18) 82(17) 11(16) AD(15) E4(15) 4F(13) 6A(13)
9	1/ 2	7B(44) E2(30) 11(27) DE(23) A4(20) 66(19) E9(18) 64(17) E6(17) 6F(16) 16(15) 2F(14)
10	1/ 1	01(0) 02(0) 03(0) 04(0) 05(0) 06(0) 07(0) 08(0) 09(0) 0A(0) 0B(0) 0C(0)

KEY FOUND! [AE:5B:7F:3A:03:D0:AF:9B:F6:8D:A5:E2:C7]

Decrypted correctly: 100%

Crack the WEP password using the Korek mode with the command below

```
crypto@crypto:~/Documents$ aircrack-ng -w password.lst wpa2.cap
Opening wpa2.cap
Read 5 packets.
```

#	BSSID	ESSID	Encryption
1	00:14:6C:7E:40:80	Harkonen	WPA (1 handshake)

Choosing first network as target.

Opening wpa2.cap
Reading packets, please wait...

Aircrack-ng 1.2 rc4

[00:00:00] 4/233 keys tested (377.64 k/s)

Crack the WPA2 password

Q1: What is the Vulnerability of WPA2 Personal

WPA2 Personal, while a significant improvement over older protocols, has a major weakness: the KRACK (Key Reinstallation Attack) vulnerability. This allowed attackers to potentially decrypt traffic transmitted on the network. Thankfully, patches were released to address KRACK.

Q2: How WPA3 addresses WPA2 shortcomings?

WPA3 fixes the KRACK vulnerability and offers other improvements over WPA2 Personal:

Stronger encryption: WPA3 uses GCMP-256 encryption compared to WPA2's 128-bit encryption, making it much harder to crack.

Dictionary password protection: WPA3 hinders brute-force attacks by only allowing one offline password guess.

Improved handshake: WPA3 uses a more secure handshake process (Dragonfly) to make it difficult for attackers to steal passwords.

Q3: Is there any possible attacks against WPA3

While WPA3 is more secure, researchers have identified potential vulnerabilities:

Downgrade attacks: An attacker might try to trick your device into using a less secure WPA2 connection.

Side-channel attacks (theoretical): In theory, complex attacks might exploit weaknesses in how devices handle WPA3.

Limited device support: Not all devices are compatible with WPA3 yet, potentially creating security gaps in a network.

Q1: How to capture the Four-Way Handshake?

Capturing a four-way handshake is a complex process that typically involves specialized tools and a controlled environment. Here's a general overview

Set up a controlled environment: Create a test network isolated from any real devices to ensure you're not capturing sensitive information.

Put your wireless adapter in monitor mode: This allows capturing all wireless traffic, not just what's addressed to your device. Specific instructions depend on your operating system and tools.

Use a packet capture tool: Tools like Wireshark are commonly used to capture network traffic.

Simulate a connection attempt: There are tools and techniques to initiate a connection to a WPA2 access point (AP) to trigger the handshake.

Q2: What is the vulnerability of WEP used in this cracking?

WEP has several vulnerabilities that make it relatively easy to crack its encryption:

Static key: WEP uses the same encryption key for all communication, making it vulnerable to attacks that can capture this key.

Weak key integrity check: The checksum used in WEP (CRC-32) is not strong enough to prevent attackers from manipulating packets and potentially decrypting them.

Initialization Vector (IV) reuse: WEP reuses the IV (a random value used in encryption) for multiple packets, making it predictable and exploitable.

These vulnerabilities render WEP unsuitable for modern security needs. Do not use WEP for any sensitive communication.

Q3: What is the vulnerability of WPA2 used in this cracking.

WPA2, while significantly stronger than WEP, has a vulnerability called KRACK (Key Reinstallation Attack). This vulnerability allowed attackers to potentially decrypt traffic under certain conditions. However, this vulnerability has been patched in most devices by now.

Q4: What have you learnt in this lab?

Importance of strong encryption: Use WPA3 or the strongest available encryption protocol on your wireless network.

Patching vulnerabilities: Keep your devices and routers updated with the latest security patches to address known vulnerabilities.

Password security: Use strong, unique passwords for your Wi-Fi network and avoid using the same password for multiple accounts.

LAB SUMMARY:

1. So, in the lab, we've been learning about AES encryption. It's pretty fascinating stuff. Basically, it's a way to keep data safe by scrambling it using a key. I think I've got a good handle on how it works now.
2. We've also been getting to grips with aircrack-ng, this set of tools for checking out Wi-Fi security. It's kind of like a toolbox for hackers, but in a legal and ethical way, you know? I'm starting to feel more confident using it.
3. One of the things we've been doing is trying to crack WEP passwords using something called KOREK mode. It's a bit like trying to pick a lock but for Wi-Fi networks. It's tricky, but I think I'm starting to understand the process.
4. Then there's WPA2, another level of Wi-Fi security. We've been looking at how passwords can be cracked with it and learning about this thing called the four-way handshake. It's all about how devices communicate securely with a Wi-Fi network. It's complex, but I'm slowly getting there.
5. And finally, we've been diving into WPA3, the latest and supposedly greatest in Wi-Fi security. It's meant to be a big improvement over WPA2, with better encryption and protection

against attacks. But, of course, there are always potential threats with new technologies, so it's important to understand those too. It's all pretty cutting-edge stuffs.

References:-

1. lab manual
2. What's the difference between WPA2 and WPA3 by William G. Wong
3. Chat gpt to paraphrase my own sentences.